```
yl15687_nyu_edu@nyu-dataproc-m:~$ cd ~

hadoop jar /usr/lib/hadoop/hadoop-streaming-3.3.6.jar \
  -D mapreduce.job.reduces=1 \
  -files mapper2.py,reducer2.py \
  -mapper "python3 mapper2.py" \
  -reducer "python3 reducer2.py" \
  -input /user/$USER/hw1/input.txt \
  -output /user/$USER/hw1/q2_output
packageJobJar: [] [/usr/lib/hadoop/hadoop-streaming-3.3.6.jar] /tmp/streamjob14993544150078l853.jar tmpDir=null
2026-09-24 05:31:36,714 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at nyu-dataproc-m.c.hpc-dataproc-19b8.internal./192.168.1.70:8032
2026-09-24 05:31:36,903 INFO client.AHSProxy: Connecting to Application History server at nyu-dataproc-m.c.hpc-dataproc-19b8.internal./192.168.1.70:10200
2026-09-24 05:31:37,657 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at nyu-dataproc-m.c.hpc-dataproc-19b8.internal./192.168.1.70:8032
2026-09-24 05:31:37,657 INFO client.AHSProxy: Connecting to Application History server at nyu-dataproc-m.c.hpc-dataproc-19b8.internal./192.168.1.70:10200
2026-09-24 05:31:38,040 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/yl15687_nyu_edu/.staging/job_1788279331828_0992
2026-09-24 05:31:38,506 INFO mapred.FileInputFormat: Total input files to process : 1
2026-09-24 05:31:38,574 INFO mapreduce.JobSubmitter: number of splits:6
2026-09-24 05:31:38,763 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788279331828_0992
2026-09-24 05:31:38,763 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-24 05:31:39,040 INFO conf.Configuration: resource-types.xml not found
2026-09-24 05:31:39,041 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-24 05:31:39,122 INFO impl.YarnClientImpl: Submitted application application_1788279331828_0992
2026-09-24 05:31:39,156 INFO mapreduce.Job: The url to track the job: http://nyu-dataproc-m.c.hpc-dataproc-19b8.internal.:8088/proxy/application_1788279331828_0992/
2026-09-24 05:31:39,157 INFO mapreduce.Job: Running job: job_1788279331828_0992
2026-09-24 05:31:50,288 INFO mapreduce.Job: Job job_1788279331828_0992 running in uber mode : false
2026-09-24 05:31:50,289 INFO mapreduce.Job:  map 0% reduce 0%
2026-09-24 05:32:02,419 INFO mapreduce.Job:  map 100% reduce 0%
2026-09-24 05:32:11,472 INFO mapreduce.Job:  map 100% reduce 100%
2026-09-24 05:32:12,487 INFO mapreduce.Job: Job job_1788279331828_0992 completed successfully
2026-09-24 05:32:12,596 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=826232
                FILE: Number of bytes written=3706757
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=2037580
                HDFS: Number of bytes written=132
                HDFS: Number of read operations=23
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=3
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=6
                Launched reduce tasks=1
```

```
yl15687_nyu_edu@nyu-dataproc-m:~$ hdfs dfs -ls /user/$USER/hw1/q2_output
Found 2 items
-rw-r--r--   1 yl15687_nyu_edu yl15687_nyu_edu          0 2026-09-24 05:32 /user/yl15687_nyu_edu/hw1/q2_output/_SUCCESS
-rw-r--r--   1 yl15687_nyu_edu yl15687_nyu_edu        132 2026-09-24 05:32 /user/yl15687_nyu_edu/hw1/q2_output/part-00000
yl15687_nyu_edu@nyu-dataproc-m:~$ hdfs dfs -cat /user/$USER/hw1/q2_output/part-00000
Word    Length  Frequency
indiscriminately        16      2
impracticability        16      1
incomprehensible        16      1
unenforceability        16      1
unreasonableness        16      1
```